

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech (ME) Examination

MAY 2018

ME421 Automobile Engineering

Date: 09.05.2018, Wenesday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following short questions below. [07]

- a. What are the main forces which oppose the motion of a vehicle?
- b. Enumerate the specifications which should be considered while purchasing an automobile.
- c. What do you mean by chassis?
- d. What are functions of frame?
- e. Why are “mag wheels”, so called?
- f. Where is clutch located?
- g. If a gearbox is noisy what items should be checked.

Q-2.(a) Enlist the defects in chassis frame. Define the following terms. [04]

(i) Wheel base (ii) Front overhang (iii) Rear overhang

(b) Explain construction and working of cone clutch. [05]

(c) What is an overdrive? Explain its construction and working. [05]

Q-3.(a) What is self-propelled vehicle? Give significance of “power to weight” ratio on an automobile’s performance. [04]

Q-3.(b) Answer any two. [10]

- i. What are functions of tyre? What is tyre rotation and its purpose?
- ii. Explain coil ignition with neat sketch.
- iii. Write a brief note on Bharat Stage emission standards.

SECTION – II

Q - 4 Answer the following short questions below. [07]

- a. What are advantages of synchromesh mechanism?
- b. Why do vehicles pull to one side?

- c. What is slip joint?
- d. What are various universal joint in use?
- e. What is necessity of propeller shaft?
- f. In which vehicle live axle is in front?
- g. What is unsprung weight?

Q-5.(a) Describe briefly the following basic suspension movements of a car. **[04]**

(i) bouncing (ii) rolling

(b) Explain construction and working of disc brake, also enlist its advantages and disadvantages. **[05]**

(c) Explain construction and working of differential with neat sketch. **[05]**

Q-6.(a) Give briefly the description of following steering parts. **[04]**

(i) steering wheel (ii) steering column (iii) steering shaft (iv) steering gearbox

Q-6 (b) **Answer any two.** **[10]**

- i. Explain construction and working of fluid coupling.
- ii. Explain trouble shooting for (i) propeller shaft shakes (ii) noisy running of propeller shaft.
- iii. Considering all factors prepare a template of auto inspection card for an automotive garage.
